Miniproject 1
Pollutant dispersion: deterministic and stochastic models

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Overview

Pollutant dispersion in the atmosphere, ocean, or groundwater is a classical example of a random transport process that can be modeled using diffusion and advection mechanisms.

Figure 1: Dispersion of particulate pollutants, taken from `https://www.oca.eu`

In this project, we will model and simulate the dispersion of pollutants in a fluid medium (such as air or water) using both *deterministic* and *stochastic* formulations. You will explore how diffusion and advection interact to transport pollutants and compare two complementary perspectives:

(i) The macroscopic (Eulerian) viewpoint, described by the advection–diffusion partial differential equation (PDE);

(ii) The microscopic (Lagrangian) viewpoint, described by stochastic differential equations (SDEs) for individual particle motion.

This project connects mathematical modeling with numerical simulation and highlights how Brownian motion can be applied to describe pollutant dispersion in a realistic physical setting. By completing this project, you will:

- Understand the relationship between the advection–diffusion equation and its stochastic formulation;
- Implement the Euler–Maruyama method to simulate stochastic particle trajectories;
- Estimate the pollutant concentration field from particle data without solving the PDE;
- Interpret the physical meaning of advection, diffusion, and source terms;
- Visualize and discuss the influence of key parameters such as the velocity field, diffusion coefficient, and source strength.

Deterministic model

For simplicity, we consider a two-dimensional (2D) model in place of the full three-dimensional (3D) problem.

Let $C(\mathbf{x}, t)$ denote the concentration of pollutant particles at position $\mathbf{x} = (x, y) \in \Omega \subset \mathbb{R}^2$ and time t. In the absence of any pollutant source, the concentration field $C(\mathbf{x}, t)$ evolves according to the advection–diffusion equation

$$\frac{\partial C}{\partial t} + \nabla \cdot (\mathbf{u}C) = D\nabla^2 C. \tag{0.1}$$

Here Ω denotes the domain (medium) of interest (here a rectangle) where the pollution is analyzed, $\mathbf{u}(\mathbf{x}, t)$ is the prescribed velocity field, $D > 0$ is the molecular diffusion coefficient, $\nabla\cdot$ is divergence operator, and ∇^2 is the Laplacian operator. Equation (0.1) describes the conservation of mass of a scalar quantity subject to *advection* by the velocity field $\mathbf{u}$ and *diffusion* due to molecular agitation.

In large environmental domains Ω, an absorbing (zero-concentration) condition $C = 0$ on $\partial\Omega$ (the boundary of Ω) is typically assumed far from the source, since the pollutant concentration decays to negligible levels at large distances.

Note: The goal of this project is not to solve the PDE (0.1) directly; instead, we focus on simulating its stochastic counterpart.

Stochastic model

An equivalent microscopic description of the same process is obtained by considering the stochastic motion of *individual* pollutant particles. Each particle trajectory $\mathbf{X}(t)$ satisfies the Itô stochastic differential equation (SDE)

$$d\mathbf{X}(t) = \mathbf{u}(\mathbf{X}(t), t)\, dt + \sqrt{2D}\, d\mathbf{B}(t), \tag{0.2}$$

where $\mathbf{B}(t)$ is a multi-dimensional Wiener process (standard Brownian motion). The first term represents deterministic *advection* by the velocity field, and the second term represents random *diffusion* with intensity proportional to D.

Connection between two models

Let $f(\mathbf{x}, t)$ denote the probability density function (pdf) of the random variable $\mathbf{X}(t)$. Then $f(\mathbf{x}, t)$ satisfies the *Fokker–Planck equation* [1],

$$\frac{\partial f}{\partial t} + \nabla \cdot (\mathbf{u} f) = D \nabla^2 f,$$

which is identical in form to (0.1). Thus, the two models describe the same underlying physics from two complementary viewpoints:

$$C(\mathbf{x}, t) = f(\mathbf{x}, t) = \int_\Omega \delta(\mathbf{x} - \mathbf{y}) f(\mathbf{y}, t)\, d\mathbf{y} = \mathbb{E}_f\left[\delta\big(\mathbf{x} - \mathbf{X}(t)\big)\right], \tag{0.3}$$

where δ is the Dirac delta function[1]. According to (0.3), the pollutant concentration field $C(\mathbf{x}, t)$ is given by the expectation $\mathbb{E}_f\left[\delta\big(\mathbf{x} - \mathbf{X}(t)\big)\right]$ Therefore, estimating this expectation yields an estimate of the concentration field. We approximate this expectation using a Monte Carlo method. To this end, we first replace the Dirac delta distribution $\delta(\mathbf{x})$ with a smooth approximation. The Dirac delta is not a classical function but a distribution that is infinite at $\mathbf{x} = \mathbf{0}$ and zero elsewhere. In 2D, we approximate it by a Gaussian kernel (a normal distribution with mean $\mathbf{0}$ and standard deviation ε):

$$\delta_\varepsilon(\mathbf{x}) = \frac{1}{2\pi\varepsilon^2} \exp\left(-\frac{\mathbf{x}^T\mathbf{x}}{2\varepsilon^2}\right), \quad \mathbf{x} \in \mathbb{R}^2. \tag{0.4}$$

Here $\mathbf{x}^T\mathbf{x}$ is the inner product of 2D vector (point) $\mathbf{x}$ with itself. As $\varepsilon \to 0$, the function δ_ε converges to the Dirac delta distribution δ in the sense of distributions. See Figure 2.

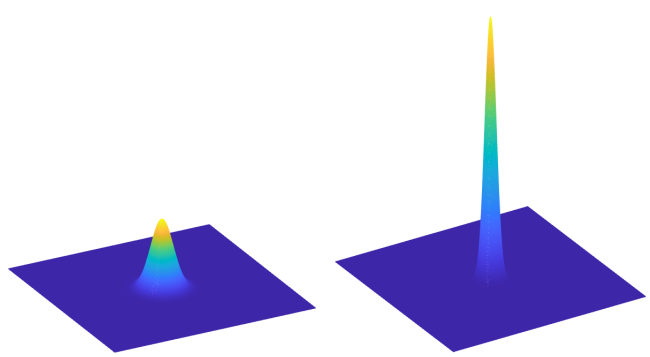

Figure 2: Gaussian kernel (0.4) with $\varepsilon = 0.2$ (left) and $\varepsilon = 0.1$ (right).

Now we replace $\mathbb{E}_f\left[\delta\big(\mathbf{x} - \mathbf{X}(t)\big)\right]$ by $\mathbb{E}_f\left[\delta_\varepsilon\big(\mathbf{x} - \mathbf{X}(t)\big)\right]$ with a small ε, and estimate it using Monte Carlo via

$$C(\mathbf{x}, t) \approx \frac{1}{N} \sum_{k=1}^{N} \delta_\varepsilon(\mathbf{x} - \mathbf{X}^k(t)), \tag{0.5}$$

[1] Any continuous function f can be represented as $f(\mathbf{x}) = \int \delta(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) d\mathbf{y}$ where δ is the Dirac delta function. In addition, if $\mathbf{X}$ has a pdf f then the expectation of random variable $g(\mathbf{X})$ is defined by $\mathbb{E}_f(g(\mathbf{X})) = \int g(\mathbf{y}) f(\mathbf{y}) d\mathbf{y}$.

provided that we have N realizations $\mathbf{X}^1, \mathbf{X}^2, \ldots, \mathbf{X}^N$ of $\mathbf{X}$ at time t. The parameter $\varepsilon > 0$ controls the spatial resolution of the reconstruction. Note that we must use $\mathbf{x} - \mathbf{X}^k(t)$ instead of $\mathbf{x}$ in kernel (0.4).

Project Tasks

Task 1: Numerical simulation (SDE approach)

(a) Implement the *Euler–Maruyama* method in Python to numerically integrate the SDE (0.2) over the time interval $t \in [0, T]$ for N independent particles. Starting with an initial position $\mathbf{X}_0$, the update rule for each particle position is given by

$$\mathbf{X}_{n+1} = \mathbf{X}_n + \mathbf{u}\, h + \sqrt{2D\, h}\, \mathbf{Z}_n, \quad \mathbf{Z}_n \sim \mathcal{N}(\mathbf{0}, I_2),$$

where I_2 is the 2×2 identity matrix and $h = t_{n+1} - t_n$ is the chosen time step. Here, $\mathbf{X}_n$ represents the numerical approximation of the particle position $\mathbf{X}(t_n)$.

(b) Consider the two-dimensional computational domain $\Omega = [0, 25] \times [-5, 5]$ (rectangle), and initialize all $N = 2000$ particles at the origin $\mathbf{X}_0 = (0, 0)$. Use the constant parameters $D = 0.02$ (diffusion coefficient) and $\mathbf{u} = [0.3, 0]$ (advective velocity in x-direction), with a final simulation time $T = 60$ seconds and time step $h = 0.1$ seconds.

Plot the particle positions (the "particle cloud") at four distinct time levels: $t = 15$, 30, 45, and 60 seconds. Each time level should be presented in a separate subplot within a single figure for visual comparison.

Use the `matplotlib.pyplot` library and the `scatter` function to visualize the simulated particle clouds in two dimensions. Use the same x and y axes limits for all four plots.

Task 2: Concentration field estimation

(a) Estimate the pollutant concentration field using equation (0.5) with a smoothing parameter $\varepsilon = 0.1$. Evaluate $C(\mathbf{x}, t)$ on a uniform spatial fine grid (e.g. a 200×100 grid) covering the domain $\Omega = [0, 25] \times [-5, 5]$.

(b) Visualize the concentration field by generating 2D contour surface plots of $C(x, y, t)$ at four time levels $t = 15, 30, 45$, and 60 seconds. Use a consistent color scale across all plots, where low concentrations appear light (e.g., white) and high concentrations appear dark (e.g., red). The `contourf` function from the `matplotlib.pyplot` library is recommended for this task. Arrange the results in a 2×2 subplot layout with a shared horizontal colorbar for clarity and the same axis limits in x and y directions.

(c) Analyze and discuss the simulation results, focusing on how advection (mean flow $\mathbf{u}$) and diffusion (random motion) jointly affect the evolution and shape of the pollutant plume over time. You do not need to include additional figures, but you should run your code with different parameter values and summarize your observations in the report.

Task 3: Including a source term

In realistic pollution scenarios, pollutants are often continuously or intermittently released into the environment by an external source such as an exhaust pipe (see Fig. 1), a leaking valve, or a localized point discharge. Here we consider a point source located at position $\mathbf{x}_s$, which continuously emits pollutants into the medium.

In the deterministic formulation, this is modeled by adding the source term $S(\mathbf{x}, t) = Q_0\, \delta(\mathbf{x} - \mathbf{x}_s)$ to the right-hand side of the advection–diffusion equation (0.1) where Q_0 denotes the constant emission rate at the source position $\mathbf{x}_s$, and δ is the Dirac delta distribution.

In the stochastic framework, this corresponds to the continuous injection of new particles into the system over time. We assume that $N(t)$ is the total number of particles at time t. Initially, we start with $N(0) = 0$ particles, and at each time step new particles are generated at the source location $\mathbf{x}_s$. The total number of particles therefore increases over time as the source remains active. Specifically:

- At each discrete time t_k, new particles are introduced at the source position $\mathbf{x}_s$.
- If we assume that the average emission rate is Q particles per second, then the number of new particles added at each time step is Qh (it is better to say $\lfloor Qh \rfloor$ because it should be an integer), where h is the time step size.
- Each particle evolves independently according to the SDE (0.2), and its trajectory is obtained using the Euler–Maruyama method.

The concentration field at any time t can then be estimated from all current particle positions using equation (0.5), where the number of active particles N is replaced by $N(t)$. Your tasks for this part are listed as below:

(a) Extend your Python code from Task 1 to include a continuous point source. Use the two-dimensional domain $\Omega = [0, 25] \times [-5, 5]$, with source location $\mathbf{x}_s = (0, 0)$, emission rate $Q = 100$ particles per second, diffusion coefficient $D = 0.02$, and velocity field $\mathbf{u} = [0.3, 0]$. Initialize with no particles ($N(0) = 0$) and simulate up to a final time of $T = 60$ seconds. Visualize the resulting particle clouds at $t = 15, 30, 45$, and 60 seconds.

(b) Repeat the concentration field estimation procedure from Task 2 for this new problem. Compute and plot $C(x, y, t)$ at the same time levels using the same visualization format.

(c) Compare and discuss the results obtained here with those from Tasks 1 and 2. Comment on the differences in particle distribution and concentration patterns, particularly the effects of continuous emission versus the initial single-release scenario.

Task 4: Discussion and report

Write a report (approximately 4–6 pages) and submit is as a **PDF file**. Assume that the reader does not have access to the project statement. The report should summarizes your modeling, implementation, and simulation results. It should be well-organized and clearly written, and should include the following components:

- A concise explanation of the conceptual link between the deterministic (PDE) and stochastic (SDE) formulations of pollutant transport, highlighting how both describe the same physical process from different perspectives.
- The main steps, assumptions, and results obtained in each task, accompanied by visualizations of your simulations. Include plots of particle clouds and estimated concentration fields with appropriate captions, axis labels, and color bars. Ensure that figure styles, scales, and dimensions are consistent throughout.
- A discussion of the physical interpretation and roles of the model parameters D, $\mathbf{u}$, and Q, explaining how each influences the evolution of the pollutant distribution.
- A reflective comparison of the deterministic and stochastic approaches, addressing their respective advantages, limitations.
- Include your fully commented Python scripts in an Appendix. The Appendix is not counted for the 4–6 page limit.

References

[1] H. Risken, The Fokker–Planck Equation: Methods of Solution and Applications, Springer, 2nd Edition, 3rd Printing (1996).